

KINGSTONE, Australia

WMO: 958070

Lat: 35.7114S

Lon: 137.5231E

Elev: 7

StdP: 101.24

Time Zone: 9.50 (AUA)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.5	2.8	0.4	3.9	6.0	1.6	4.2	6.5	11.9	12.6	10.9	12.9	0.8	260	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.0	34.1	18.8	31.3	18.5	28.7	17.9	20.7	28.3	19.8	27.6	18.9	26.0	5.9	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.5	22.3	17.4	12.4	21.4	16.4	11.7	20.8	60.0	28.0	56.7	27.7	53.8	26.3	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.8	8.9	-0.5	39.7	0.8	2.3	-1.1	41.4	-1.6	42.7	-2.0	44.0	-2.6	45.7		
			-0.8	22.5	0.9	1.2	-1.4	23.4	-1.9	24.1	-2.4	24.8	-3.0	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	19.6	19.5	17.6	15.2	12.9	11.2	10.5	10.6	11.7	13.3	15.8	17.6
	DBStd	4.18	3.27	3.09	2.95	2.45	2.19	2.00	1.99	1.98	2.31	2.71	3.24	3.17
	HDD10.0	53	0	0	0	0	2	11	17	16	6	1	0	0
	HDD18.3	1555	20	17	50	101	168	216	242	240	199	160	91	52
	CDD10.0	1732	297	267	236	155	93	45	34	34	58	103	175	236
	CDD18.3	192	58	51	27	6	1	0	0	0	1	3	16	30
	CDH23.3	2349	676	504	316	76	3	0	0	0	2	76	271	428
	CDH26.7	1018	329	228	127	14	0	0	0	0	0	21	108	191
Wind	WSAvg	4.2	5.3	5.1	4.4	3.5	3.5	3.5	3.9	3.8	4.0	4.2	4.6	4.8
Precipitation	PrecAvg	485	16	20	21	31	50	77	74	63	51	34	23	23
	PrecMax	693	64	71	107	68	84	131	132	100	128	101	60	61
	PrecMin	328	2	0	1	3	11	23	29	14	16	1	4	4
	PrecStd	94	14	20	22	19	19	28	25	26	28	26	14	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.8	36.7	34.0	28.7	23.2	18.4	17.4	19.6	22.8	29.6	33.4	36.4
		MCWB	20.0	20.3	18.7	16.9	15.1	14.1	13.0	13.7	14.6	16.2	18.3	18.4
	2%	DB	33.9	32.7	30.3	25.2	20.7	17.0	16.2	17.7	20.4	25.2	29.9	31.6
		MCWB	19.1	18.9	18.3	16.2	15.1	13.5	12.5	12.9	14.5	15.5	17.4	17.6
	5%	DB	30.3	29.3	26.7	23.2	19.0	16.1	15.4	16.3	18.6	22.1	26.5	28.1
		MCWB	18.8	18.9	17.8	16.1	14.6	12.9	12.1	12.4	13.6	15.1	17.0	17.4
	10%	DB	26.6	25.9	23.6	21.2	17.8	15.3	14.6	15.3	17.2	19.7	23.1	24.7
		MCWB	18.2	18.2	17.1	15.6	14.1	12.4	11.7	11.8	13.1	14.3	16.1	16.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.1	22.5	21.0	18.9	17.4	15.2	14.2	14.8	16.3	17.8	20.1	20.6
		MCDB	31.3	30.1	28.1	23.2	19.6	16.9	16.0	17.4	20.2	24.2	29.5	27.5
	2%	WB	20.7	20.7	19.7	17.8	16.1	14.2	13.2	13.6	15.2	16.6	18.5	19.2
		MCDB	29.6	27.0	26.3	22.4	18.8	16.1	15.3	16.4	18.7	22.7	26.9	27.5
	5%	WB	19.6	19.7	18.6	17.0	15.3	13.5	12.6	12.8	14.3	15.6	17.5	18.1
		MCDB	27.8	27.0	24.7	21.5	18.0	15.5	14.7	15.7	17.7	21.0	24.8	26.0
	10%	WB	18.4	18.7	17.6	16.1	14.5	12.8	12.0	12.1	13.4	14.6	16.3	17.1
		MCDB	25.9	25.0	22.8	20.3	17.3	14.9	14.2	14.8	16.6	19.2	22.3	23.9
Mean Daily Temperature Range	5% DB	MDBR	12.0	11.3	11.6	11.6	9.4	8.2	8.1	9.2	10.3	12.0	12.2	12.6
		MCDBR	20.2	18.5	17.9	16.0	11.9	9.2	8.4	10.7	13.0	18.0	19.3	20.0
		MCWBR	7.0	6.6	7.6	8.3	7.6	6.4	5.7	7.1	8.0	9.5	8.2	7.5
	5% WB	MCDBR	15.9	14.4	14.3	13.2	10.0	8.5	8.0	10.0	11.5	15.2	16.3	15.6
		MCWBR	6.5	5.8	6.9	7.5	6.8	6.3	5.9	7.1	7.4	8.7	7.6	6.8
Clear-Sky Solar Irradiance	taub	0.344	0.342	0.324	0.319	0.312	0.306	0.305	0.314	0.338	0.344	0.343	0.339	
	taud	2.564	2.582	2.631	2.628	2.625	2.626	2.622	2.583	2.508	2.492	2.523	2.561	
	Ebn at Noon	992	972	954	900	850	827	849	891	922	958	985	1001	
	Edh at Noon	107	101	89	79	70	65	69	81	99	109	110	108	
All-Sky Solar Radiation	RadAvg	7.32	6.28	5.05	3.65	2.52	2.08	2.26	3.13	4.35	5.60	6.69	7.18	
	RadStd	0.35	0.29	0.19	0.23	0.12	0.12	0.11	0.18	0.34	0.38	0.31	0.33	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+129	+53
(47)	Regional (23 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-98	+161	+70

Nomenclature: See separate page